



## OCZ Vector 180

Toshiba just let loose their new flagship 6Gbps SSD – the OCZ Vector 180 featuring a Barefoot 3 controller <http://dgit.in/BareOCZ>



## Kurganets BMP-25

A new armoured prototype Russian infantry fighting vehicle has been spotted at a parade rehearsal <http://dgit.in/ArmaRus>

Geek life

the clunky offering of its first prototype, the Developer Kit 1 (or DK1), which was followed by improved versions such as Crystal Cove and DK2, with better technical specifications and optimized design and ergonomics. In the heart of the device is housed an ARM processor and collection of chips used to control the LEDs and the “Adjacent Reality Tracker”—which in essence is a magnetometer, a gyroscope and an accelerometer, all used for achieving accurate head tracking in terms of pitch, roll, and yaw.

For video input, the Rift uses an HDMI cable, while a standard USB 2.0 cable is used to power the device and interface it with a PC for updating firmware, analyzing stats and other geek-related uses. The headset comes with a built in microUSB port, giving you the option to potentially connect a controller, a USB headset, or even a USB-powered neck massager for those extended Saturday night gaming sessions (Goodbye, social life!). The headset is strapped onto your head via vertical and horizontal straps, with the input cables attached to the uppermost one.

The latest version—aka Crescent Bay—offers reduced weight, an inbuilt audio processor (previous versions required the developer to implement their own audio module), and tracking LEDs in the back of the headset that now allow for 360-degree head tracking. This version also supports movement tracking at a very high polling rate of 1000 Hz for truly smooth VR head-turning. All of this is output on a pair of 1920 x 1080 HD AMOLED displays

(same as those used in Samsung’s Galaxy Note 3) at a smooth refresh rate of 90 Hz, which is a major factor for minimizing motion sickness.

One major improvement that Crescent Bay can boast of over previous Rift versions is the integration of Head-Related Transfer Function technology (HRTF) in the official Oculus Audio SDK. In their words, “HRTFs simulate the changes to a sound when it reaches your head from a point in space [...] by referencing data that represents changes that would happen to a sound coming from that direction.” In simpler terms, the technology can determine the sound to be generated in advance, thus playing it at a lower latency for a smoother 3D audio experience. Unfortunately, despite such progress, Crescent Bay will not be the consumer edition of the Rift. Expect the final version to have additional improvements and possibly even a wireless version.

## HTC Vive

While Oculus seems set to make the biggest dent in the upcoming VR market, it is in no way going to do so unopposed. By collaborating with the likes of PC gaming giants Valve, HTC have shown that they mean business with the introduction of the HTC Vive. While they may have taken heavy inspiration from Crescent Bay in terms of design and specs, HTC and Valve have more than a few tricks

up their sleeve which should give the Vive an edge over its competition.

For one, HTC boast that the headset will offer unprecedentedly low levels of latency as well as super smooth 360-degree head movement, along with the ability the walk around in a 15 x 15 square feet area,



PC gamers will be hoping Valve’s reputation pays off

making for exceptionally immersive VR with unparalleled levels of maneuvering freedom. However, the one exceptional (alleged) selling point of HTC’s Vive is their claim that the final version will never induce motion sickness. In terms of VR headsets, this is quite a tall claim to make as quite a few first-time users of other VR headset complain about queasiness and minor headaches brought about by low framerates and high input latency. Furthermore, the device will be powered by SteamVR, which should be a game-making offering backed by Steam’s 125 million strong user base and Valve’s esteemed reputation for making excellent gaming experiences.

Specifications-wise, the Vive offers two 1200 x 1080p HD displays, with a maximum refresh rate of 90Hz. The integrated accelerometer, gyroscope and laser positioning system allows for highly accurate head tracking: up to one-tenth of a degree in any direction. The “contextually aware controller” (better known to most as Valve’s Steam controller) it comes with should allow you to intuitively move about and interact with objects in the virtual world.

Unlike Oculus Rift, HTC and Valve have a proper release date set at Christmas 2015. In fact, should HTC beat Oculus and others to the punch, they have a good chance of capturing a sizeable portion of VR headset sales.



The device that started it all